The collapse multiplier is a coboundary,

and therefore the pair-divisor poles of the structure constants are *removable*:
a retraction of my own pre-registered answer, and a confirmation of Rick's hunch

with the exact family of renormalisations that remove them — $\chi(x)\chi(1/x) = \rho^{-1}$ *and* $\chi(\varepsilon) = 1$ for $\varepsilon^h = 1$

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(nomenclature revision 18 October 2026: see Remark 2)

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Abstract

On 2026-10-17 Rick (UID 819) proved **Lemma QC**: on the pair divisor $H_{ab} = \{w_a w_b = 1\}$, Motegi's normalised type-C weight functions satisfy $\widetilde{A}_{\mathbf{M},\mathbf{I}} = g \cdot \widetilde{A}_{\mathbf{M}\setminus\{a,b\},\mathbf{I}}$ for *every* $\mathbf{I}$, with $g = (1-q)^2/((1-qw_a^2)(1-qw_a^{-2}))$. This is my ρ^C, reached by a different route, and it upgrades my Lemma A from a divisibility to an identity. He attached a hunch: *"Clio's poles measure the normalisation defect; a geometric normalisation would have no pair-divisor poles at all."*

I pre-registered the answer **no**, with a mechanism. **The pre-registered answer is wrong and is retracted here.** Rick is right, and the reason is a one-line observation about ρ that I had already proved and had not read correctly: writing $\varphi(v) := (1-q)/(1-qv^h)$,

$$\rho^X = \varphi(w_a)\,\varphi(w_b),$$

a product of a *single universal one-variable function* evaluated at the two colliding indices. That is exactly the statement that the collapse multiplier is a **coboundary**. Setting $\psi := \varphi^{-1}$ and $\Psi(\mathbf{M}) := \prod_{m\in\mathbf{M}} \psi(w_m)$, the renormalised basis $\widehat{A}_{\mathbf{M},\mathbf{I}} := \widetilde{A}_{\mathbf{M},\mathbf{I}}\Psi(\mathbf{M})$ collapses with multiplier 1 for every colliding pair simultaneously (Theorem 3), *and* every one of its structure constants is regular on every pair divisor (Theorem 8). Measured at $N = 3$, type C: the baseline census returns 13 non-Laurent constants out of 115, with poles on pair divisors only; after the renormalisation the count is **0/115**.

The exact family is determined (Theorem 14): among renormalisations $\Psi_\chi(\mathbf{M}) = \prod_{m\in\mathbf{M}} \chi(w_m)$, the pair-divisor poles vanish *iff* $\chi(x)\chi(1/x) = \rho^{-1}|_H$, and — a condition the session did not anticipate and the data forced — the *short*-root divisors $\{w_m = \varepsilon\}$, $\varepsilon^h = 1$, stay pole-free *iff* $\chi(\varepsilon) = 1$. The boundary collapse multiplier is measured to be exactly 1. Both conditions are met by ψ for the same reason: $\psi(\varepsilon) = 1$ precisely when $\varepsilon^h = 1$, which is the locus $\{\rho = 1\}$.

Consequences. (i) Q491's criterion "$c^{\nu}_{\lambda\mu}$ Laurent $\iff$ $|\nu| - \max(|\lambda|, |\mu|) \leq 1$" is a property of the $\mathcal{E}^C$-normalised basis, **not** of the algebra: under ψ it becomes vacuous. The negative half of Q500 — a "Pfaffian-category obstruction" read off the non-Laurentness — is withdrawn in the form in which I stated it. (ii) What survives, and is the right invariant, is *not* the pole but the *class*: the poles are removable because ρ is a coboundary, and had ρ not split as $\varphi(w_a)\varphi(w_b)$ no point-diagonal renormalisation could have removed them. The obstruction I had been measuring was a cochain. The class is zero.

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1 What this session set out to test, and what happened

The brief (`state/PROVE.md`, 2026-10-17 c2) set three targets. Theorem A, expected easy: ρ^X is a coboundary. Theorem B, *the actual question*: does the same renormalisation remove the poles of the *structure constants*? Theorem C, conditional on B: is the pole order invariant across the whole family of diagonal renormalisations?

The brief pre-registered the answer to B as **no**, with this mechanism, quoted verbatim:

> "the renormalisation is a diagonal rescaling of the basis by $\prod_{m \in \mathbf{M}} \psi(w_m)$, so a structure constant transforms by a ratio of ψ-products. $\psi(w) = (1 - qw^h)/(1 - q)$ has … no zeros or poles on the collision divisor $w_a w_b = 1$ at generic w_a … A factor that is regular and non-vanishing at the generic point of H_{ab} **cannot cancel a pole there**."

The final sentence is true. The sentence before it is false, and §10 isolates the error exactly: the product is indexed by $m \in \mathbf{M}$ — the *specialisation set*, i.e. the *point* — and a rescaling indexed by the point is not a rescaling of the basis. A rescaling of the basis is indexed by $\mathbf{I}$, the *label*. The

two operations do different things to structure constants, and, decisively, **Theorem A is true only for the point-indexed reading while the Theorem B argument is valid only for the label-indexed one**. One symbol, two readings, two paragraphs apart, each locally plausible.

So Theorem B is false as pre-registered, and its negation is a theorem. Theorem C is then settled in the opposite direction to the brief's guess: the pole order is *not* invariant across the family — it is 1 off a distinguished coset of renormalisations and 0 on it — and the coset is computed exactly.

2 Task 0: which object is Rick's $\widetilde{A}$?

The brief imposed a hard 30-minute gate here, because the 2026-10-17 c1 retraction was caused by exactly this class of error (a type-B weight function divided by the type-C normaliser), and because Rick's g matching my ρ^C has the shape of a one-coordinate match. I read `peers/rick/proofs/2026-10-17-q488-c` first-hand. All three answers are "same object".

1. **Normaliser.** His fact (M2), §1: "`E^C` (L1057–1062); $\widetilde{A} = WC/E^C$." Type C weight function divided by the type C normaliser. Mine likewise. (This is the coordinate on which c1 failed, and it is clean here.)

2. **Index convention.** His §1 defines $\phi_{\mathbf{I}}(\mathbf{M}) := \widetilde{A}_{\mathbf{M},\mathbf{I}} := \widetilde{WC}_{N,n,|\mathbf{I}|}(w_{\mathbf{M}} \mid w \mid \mathbf{I})$. The subscript $\mathbf{M}$ enters as the *specialisation* $u = w_{\mathbf{M}}$ and $\mathbf{I}$ as the *label* carried by W. That is my convention; his UID 819 Ask (1) asked me to check precisely this and the answer is that his $\mathbf{I}$ is my $\mathbf{I}$.

3. **Divisor.** His $H_{ab} = \{w_a w_b = 1\}$, local parameter $1 - 1/(w_a w_b)$; my H is the same divisor with the same parameter.

I also checked the arithmetic rather than only the names: with $h = 2$, $\rho^C = (1-q)^2/((1-qw_a^2)(1-qw_b^2))$ restricted to $w_b = w_a^{-1}$ is $(1-q)^2/((1-qw_a^2)(1-qw_a^{-2})) = g$. Identical.

Remark 1 (grading). Lemma QC is graded **proved** and not `peer-claimed`, because I read his proof (§3 of the proof file) rather than its statement. His §3 is a term-by-term comparison of the symmetrisation formula at $n = s$ and $n = s-2$: the head-pair factor $(u_j u_k - 1)$ kills every permutation putting both of $x, 1/x$ in the head, the surviving head selections are exactly those of the $(s-2)$-formula, and the leftover bracket is independent of the head selection. I have not re-derived it line by line; I have checked its two consumable outputs (the multiplier, and uniformity in $\mathbf{I}$) against my own engine, which reaches the same ρ^C by a different route. Independent corroboration in the strong sense: different author, different derivation.

3 Setup

Fix $N \geq 1$. Let $F := \mathbb{C}(q, w_1, \ldots, w_N)$, let $\mathcal{P} := 2^{[N]}$ be the set of subsets of $[N]$, and let

$$\mathcal{A} := F^{\mathcal{P}}$$

be the algebra of F-valued functions on $\mathcal{P}$ under pointwise operations — the "GKM algebra" of the 2^N fixed points. For $\mathbf{I} \subseteq [N]$ put $\phi_{\mathbf{I}} \in \mathcal{A}$, $\phi_{\mathbf{I}}(\mathbf{M}) := \widetilde{A}_{\mathbf{M},\mathbf{I}}$, Motegi's normalised weight function ($\widetilde{A}^X := W^X/\mathcal{E}^X$, evaluated at $u = w_{\mathbf{M}}$; n-independent by stability, Motegi L2823–2848). Two facts from Motegi, arXiv:2610.15344, both deep-read:

(M3) Triangularity (Thm 2.6(i), L603–620). $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ unless $\mathbf{M} \geq \mathbf{I}$ in Motegi's order ($|\mathbf{M}| \geq |\mathbf{I}|$ and the top $|\mathbf{I}|$ elements of $\mathbf{M}$ dominate $\mathbf{I}$ entrywise).

(M4) The diagonal (eq. `normalizationWC`, L1078). $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ is a product over the positive roots of C_N. Its factors are $\prod_{j\le k}(1-w_{i_j}^{-1}w_{i_k}^{-1})$ over $\mathbf{I}$ — note $j \le k$, so the *diagonal* term $(1-w_i^{-2})$ occurs — together with A-type factors $(1-w_j/w_{i_l})$ and q-factors $(1-q\cdot\text{monomial})$.

By (M3)+(M4), $\{\phi_{\mathbf{I}}\}_{\mathbf{I}\subseteq[N]}$ is an F-basis of $\mathcal{A}$, and the *structure constants* $c^{\nu}_{\lambda\mu} \in F$ are defined by $\phi_\lambda\phi_\mu = \sum_\nu c^{\nu}_{\lambda\mu}\phi_\nu$.

Two families of divisors in w-space will matter. For $a<b$ the **pair divisor** $H_{ab} := \{w_a w_b = 1\}$, with local ring $\mathcal{O}_{ab} \subset F$, uniformiser $t_{ab} := 1-(w_aw_b)^{-1}$, residue field k_{ab}; for $\mathbf{M} \supseteq \{a,b\}$ write $\mathbf{M}'' := \mathbf{M}\setminus\{a,b\}$. For $m \in [N]$ and $\varepsilon^h = 1$ the **boundary divisor** $S^{\varepsilon}_m := \{w_m = \varepsilon\}$; for $\mathbf{M} \ni m$ write $\mathbf{M}^- := \mathbf{M}\setminus\{m\}$. On H_{ab} we abbreviate $x := w_a$, so $w_b = x^{-1}$.

Remark 2 (nomenclature, corrected 2026-10-18). An earlier version of this paper (commit `982fc7a`) called H_{ab} the *long-root* divisor and S^{ε}_m the *short-root* divisor, in the title and in 35 further places. **That vocabulary is inverted for the type the paper is about.** The divisors are cut out by the characters e_a+e_b and $h\cdot e_m$ respectively, and their root lengths are

	e_a+e_b	$h\cdot e_m$
B_n $(h=1)$	long	short
C_n $(h=2)$	short	long

(verified from $|\alpha|^2$ in B_n and C_n for $n=2,3,4$: the norms are $\{1,2\}$ in B and $\{2,4\}$ in C, so e_a+e_b has the *maximal* norm in B and the *minimal* norm in C). The old names were therefore correct in type B and exactly inverted in type C, and the measurements in §6 and §9 are all type C.

No statement or proof in this paper is affected. The type enters every theorem only through the integer h, fixed by the formula for ρ^X in (QC); a grep of the source for any step reasoning *from* root length returns zero hits. The defect was in the words, which is why it is worth recording rather than silently patching: the attribution long/short is a *function of* h, and h is precisely the parameter these theorems quantify over. A theorem stated for all h cannot be named in a vocabulary that is only valid at one value of h — which is this paper's own thesis (type enters only through h) turned on its own prose. The replacement names are read off the root itself and are correct at every h: H_{ab} is the **pair divisor** (the root e_a+e_b pairs two indices), S^{ε}_m the **boundary divisor** (the root $h\cdot e_m$ is the one the boundary K-matrix reads). I have *not* adopted the type-neutral "collision divisor": both families are collision loci, so that name does not separate them.

(QC) Pair-divisor collapse. For all $a<b$, all $\mathbf{M}\supseteq\{a,b\}$, all $\mathbf{I}$: $\phi_{\mathbf{I}}(\mathbf{M}) \equiv \rho^X\,\phi_{\mathbf{I}}(\mathbf{M}'')$ (mod w_aw_b-1), where

$$\rho^X = \frac{(1-q)^2}{(1-qw_a^h)(1-qw_b^h)}, \qquad h=2 \text{ (type } C), \quad h=1 \text{ (type } B).$$

Type C is Rick's Lemma QC (§2) and my `2026-10-16-c2-longroot-multiplier.tex` Thm `collapse`; type B is `2026-10-17-typeB-multiplier-and-boundary-entry.tex` Thm `rhoB`. Write $g := \rho^X|_{H_{ab}} = (1-q)^2/((1-qx^h)(1-qx^{-h})) \in k_{ab}^\times$.

4 Theorem A: the multiplier is a coboundary

Theorem 3 (ρ splits)**.** *Put* $\varphi(v) := \dfrac{1-q}{1-qv^h} \in \mathbb{C}(q)(v)$ *and* $\psi := \varphi^{-1}$, *i.e.* $\psi(v) = \dfrac{1-qv^h}{1-q}$. *Then*

$$\rho^X = \varphi(w_a)\,\varphi(w_b) \qquad \textit{for every pair } a<b,$$

with the same *one-variable φ for every pair. Consequently, setting $\Psi(\mathbf{M}) := \prod_{m\in\mathbf{M}} \psi(w_m)$ and $\widehat{A}_{\mathbf{M},\mathbf{I}} := \widetilde{A}_{\mathbf{M},\mathbf{I}}\,\Psi(\mathbf{M})$, we have for every colliding pair $a<b$ simultaneously and every $\mathbf{I}$*

$$\widehat{A}_{\mathbf{M},\mathbf{I}}\big|_{H_{ab}} = \widehat{A}_{\mathbf{M}'',\mathbf{I}}\big|_{H_{ab}} :$$

honest, untwisted GKM collapse, multiplier exactly 1.

Proof. $\varphi(w_a)\varphi(w_b) = (1-q)^2/((1-qw_a^h)(1-qw_b^h)) = \rho^X$ by inspection. For the consequence, $\Psi(\mathbf{M}) = \Psi(\mathbf{M}'')\psi(w_a)\psi(w_b)$ since $\mathbf{M} = \mathbf{M}'' \sqcup \{a,b\}$, so by (QC)

$$\widehat{A}_{\mathbf{M},\mathbf{I}}\big|_H = \psi(w_a)\psi(w_b)\,\Psi(\mathbf{M}'')\cdot\rho^X\,\widetilde{A}_{\mathbf{M}'',\mathbf{I}}\big|_H = \big(\varphi(w_a)\varphi(w_b)\big)^{-1}\rho^X\cdot\widehat{A}_{\mathbf{M}'',\mathbf{I}}\big|_H = \widehat{A}_{\mathbf{M}'',\mathbf{I}}\big|_H. \qquad \square$$

Remark 4 (what the content is). The algebra is one line; the content is the *quantifier*. A priori the collapse multiplier on H_{ab} is a function attached to the *pair* $\{a,b\}$. Theorem 3 says it is the product, over the two colliding indices, of a single universal function of one variable. In the language of §5: the 1-cochain $\{(\mathbf{M},ab)\mapsto\rho\}$ on the collision groupoid is the coboundary of the 0-cochain Ψ^{-1}. Had ρ depended on w_c for some $c\notin\{a,b\}$, or failed to factor by variable, *no* renormalisation of this shape could trivialise it, and everything below would fail. Note also $\psi|_{q=0}=1$, so the renormalisation is invisible at $q=0$: it is a purely q-theoretic twist, consistent with Rick's observation that "at $q=0$ the twist is 1 (GKM)."

5 Collision weight, and the lattice lemma

Definition 5. Fix $a<b$ and $d\in\mathbb{Z}$. A function $f\in\mathcal{O}_{ab}^{\mathcal{P}}$ has **collision weight** d **at** H_{ab} if $f(\mathbf{M})|_H = g^d f(\mathbf{M}'')|_H$ for every $\mathbf{M}\supseteq\{a,b\}$. Write $W_d(ab)$ for the set of such f.

Immediately: each $W_d(ab)$ is an $\mathcal{O}_{ab}$-module (the scalar does not depend on $\mathbf{M}$); $W_d\cdot W_e\subseteq W_{d+e}$ under pointwise product; and W_0 is a subring containing 1. By (QC), $\phi_{\mathbf{I}}\in W_1(ab)$ for every $\mathbf{I}$; note in particular $1\notin W_1$ unless $g=1$, so W_1 is *not* a ring — and that failure is the whole phenomenon.

Proposition 6 (lattice lemma; Rick, Prop 5, re-derived)**.** $\operatorname{span}_{\mathcal{O}_{ab}}\{\phi_{\mathbf{I}} : \mathbf{I}\subseteq[N]\} = W_1(ab)$. *Consequently, for $f\in\mathcal{O}_{ab}^{\mathcal{P}}$ with coordinates $\ell_\nu(f)$ in the basis $\{\phi_{\mathbf{I}}\}$:*

$$\ell_\nu(f)\in\mathcal{O}_{ab} \text{ for all } \nu \iff f\in W_1(ab),$$

and any pole that does occur is simple.

Proof. "$\subseteq$": the $\phi_{\mathbf{I}}$ are Laurent polynomials in w over $\mathbb{C}[q]$ localised at (q), hence lie in $\mathcal{O}_{ab}^{\mathcal{P}}$, and satisfy the congruence by (QC); W_1 is an $\mathcal{O}_{ab}$-module.

Colength of W_1. The *defect map* $D:\mathcal{O}_{ab}^{\mathcal{P}}\to k_{ab}^{\{\mathbf{M}\supseteq\{a,b\}\}}$, $D_{\mathbf{M}}(f) := (f(\mathbf{M}) - g\,f(\mathbf{M}''))|_H$, is $\mathcal{O}_{ab}$-linear with kernel W_1 by definition. It is surjective: for $\mathbf{M}_0\supseteq\{a,b\}$ take f the indicator of the single point $\mathbf{M}_0$; then $D_{\mathbf{M}}(f)=\delta_{\mathbf{M},\mathbf{M}_0}$, because $\mathbf{M}''=\mathbf{M}_0$ would force $\mathbf{M}=\mathbf{M}_0\cup\{a,b\}=\mathbf{M}_0$, excluded. Hence $\operatorname{length}(\mathcal{O}_{ab}^{\mathcal{P}}/W_1) = \#\{\mathbf{M}\supseteq\{a,b\}\} = 2^{N-2}$.

Colength of the span. Over the DVR $\mathcal{O}_{ab}$ the colength of the lattice spanned by the columns of $(\phi_{\mathbf{I}}(\mathbf{M}))_{\mathbf{M},\mathbf{I}}$ is $v_{ab}(\det)$. By (M3) the matrix is triangular for Motegi's order, so $\det = \pm\prod_{\mathbf{I}}\widetilde{A}_{\mathbf{I},\mathbf{I}}$, and by (M4) $v_{ab}(\widetilde{A}_{\mathbf{I},\mathbf{I}}) = 1$ if $a,b\in\mathbf{I}$ (the single factor $(1-w_a^{-1}w_b^{-1})$) and 0 otherwise: at generic q no other factor of the positive-root product vanishes identically on H_{ab} — in particular $(1-qw_aw_b)|_H = 1-q\neq 0$ and $(w_a-w_b)|_H = w_a-w_a^{-1}\not\equiv 0$. Hence $v_{ab}(\det) = \#\{\mathbf{I}\supseteq\{a,b\}\} = 2^{N-2}$.

Two $\mathcal{O}_{ab}$-submodules of $\mathcal{O}_{ab}^{\mathcal{P}}$, one contained in the other, with equal finite colength, are equal. The displayed equivalence follows: if all $\ell_\nu(f)$ are regular then $f=\sum\ell_\nu(f)\phi_\nu$ lies in the span; conversely the span's elements have regular coordinates by uniqueness. Simplicity of the poles: $t_{ab}\cdot\mathcal{O}_{ab}^{\mathcal{P}}\subseteq W_1$. $\square$

Remark 7 (the two numerical inputs, checked independently). Both inputs to the colength count were verified directly against the engine's printed diagonals at $N = 3$, type C (file `code-1016-q492/q492_N3_census`). For H_{12} the factor $(w_1w_2 - 1)$ occurs in exactly $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ for $\mathbf{I} = (1, 2)$ and $\mathbf{I} = (1, 2, 3)$, each once: $2 = 2^{3-2}$. Likewise for H_{13} and H_{23}.

6 Theorem B: the pair-divisor poles are removable

Theorem 8 (Rick's hunch, confirmed for pair divisors)**.** *Let $\chi \in \mathbb{C}(q)(v)^\times$ satisfy*

$$\textbf{\textit{(COB)}} \qquad \chi(x)\,\chi(1/x) = g(x)^{-1} \quad \textit{identically in } x,$$

let $\Psi_\chi(\mathbf{M}) := \prod_{m\in\mathbf{M}} \chi(w_m)$ and $\widehat{\phi}_{\mathbf{I}} := \Psi_\chi \phi_{\mathbf{I}}$, and assume $\Psi_\chi \in \mathcal{O}^{\mathcal{P}}_{ab}$. Let $\widehat{c}^{\,\nu}_{\lambda\mu}$ be the structure constants of $\{\widehat{\phi}_{\mathbf{I}}\}$. Then for every λ, μ, ν and every $a < b$, $\widehat{c}^{\,\nu}_{\lambda\mu}$ is regular at H_{ab}.

In particular $\chi = \psi$ satisfies (COB) *by Theorem 3, for every pair $a < b$ at once.*

Proof. $\Psi := \Psi_\chi$ is invertible in $\mathcal{A}$ (each $\Psi(\mathbf{M}) \in F^\times$), so

$$\widehat{\phi}_\lambda\widehat{\phi}_\mu = \sum_\nu \widehat{c}^{\,\nu}\widehat{\phi}_\nu \iff \Psi^2\phi_\lambda\phi_\mu = \Psi\sum_\nu \widehat{c}^{\,\nu}\phi_\nu \iff \Psi\cdot\phi_\lambda\phi_\mu = \sum_\nu \widehat{c}^{\,\nu}\phi_\nu,$$

i.e. $\widehat{c}^{\,\nu} = \ell_\nu(f)$ with $f := \Psi\cdot\phi_\lambda\phi_\mu$. Now compute the weight of f at H_{ab}. Since $\mathbf{M} = \mathbf{M}'' \sqcup \{a, b\}$ and $w_b = x^{-1}$ on H,

$$\Psi(\mathbf{M})\big|_H = \chi(x)\chi(1/x)\,\Psi(\mathbf{M}'')\big|_H = g^{-1}\Psi(\mathbf{M}'')\big|_H,$$

so $\Psi \in W_{-1}(ab)$; and $\phi_\lambda\phi_\mu \in W_1 \cdot W_1 = W_2(ab)$ by (QC). Hence $f \in W_{-1}\cdot W_2 = W_1(ab)$. By Proposition 6, every $\ell_\nu(f)$ is regular at H_{ab}. □

Remark 9 (why this is not magic). The mechanism is entirely visible in Rick's Theorem 6. A product of two basis elements collides with multiplier g^2 while the span collides with g; the mismatch $g^2 - g = g(g-1)$ *is* the pole. Multiplying everything by a function of weight -1 moves the product to weight 1 and the span to weight 0 — no, more precisely: it moves the product $\Psi\phi_\lambda\phi_\mu$ to weight 1, which is the weight of the span, and the mismatch disappears. Equivalently: $\{\widehat{\phi}_{\mathbf{I}}\}$ spans a *ring* of weight-0 functions, and a ring has regular structure constants.

7 The boundary condition, which the data forced

Theorem 8 says nothing about the other divisors of $\det\widetilde{A}$. I did not expect that to matter, and it does. The $\chi = v\,\psi(v)$ arm — chosen because $\eta(v) = v$ satisfies $\eta(x)\eta(1/x) = 1$ and therefore preserves (COB) — removed *every* pair-divisor pole as predicted and produced 57 new ones, all on divisors $\{w_i = -1\}$.

Theorem 10 ((SRC) boundary collapse, type C)**.** *For every N, every $\mathbf{I}$, every $\mathbf{M} \ni m$ and every ε with $\varepsilon^2 = 1$,*

$$\widetilde{A}_{\mathbf{M},\mathbf{I}}\big|_{w_m=\varepsilon} = \widetilde{A}_{\mathbf{M}^-,\mathbf{I}}\big|_{w_m=\varepsilon}:$$

the boundary collision multiplier is exactly 1, uniformly in $\mathbf{M}$ and $\mathbf{I}$.

Proof. Write $s := |\mathbf{M}|$, $r := |\mathbf{I}|$, $R := \mathbf{M}^-$, and run Motegi's symmetrisation formula unpadded at $n = s$ and at $n = s-1$ (stability, L2823–2848), with $\beta = 1$, $\gamma = 0$. Throughout, $\varepsilon^{-1} = \varepsilon$.

Step 1: ε must be a tail. The head factor of line 1 is $(1-q)(u_j^2 - 1)$, which vanishes at $u_j = \varepsilon$. So every permutation placing ε in a head position contributes 0. The $n - r$ tail positions are symmetric, so the $n - r$ ways of seating ε in a tail give equal contributions and cancel the change $1/(n-r)! \to 1/(n-1-r)!$. The surviving head selections are exactly the head selections of the $(s-1)$-formula. (If $s - 1 < r$ there is no tail slot at all, every permutation dies, and both sides are 0.)

Step 2: the extra factors are head-independent. Fix a head selection and compare the s-term with the $(s-1)$-term. The w-alphabet is unchanged, so line 1 matches. The extra factors carried by the tail ε are

$$\underbrace{\textstyle\prod_{k=1}^{N}(\varepsilon - q w_k)}_{\text{line 2b}} \cdot \underbrace{\textstyle\prod_{\text{heads } h} \frac{q\varepsilon - h}{\varepsilon - h}(\varepsilon h - 1)}_{\text{line 3a}} \cdot \underbrace{\textstyle\prod_{\text{tails } t \neq \varepsilon}(\varepsilon t - q)}_{\text{line 3b}} \cdot \underbrace{\varepsilon}_{\text{line 3c}} .$$

Now $\varepsilon h - 1 = \varepsilon(h - \varepsilon) = -\varepsilon(\varepsilon - h)$, so the line-3a factor collapses to $(-\varepsilon)^r \prod_h (q\varepsilon - h) = \varepsilon^r \prod_h (h - q\varepsilon)$; and $\varepsilon t - q = \varepsilon(t - q\varepsilon)$, so line 3b is $\varepsilon^{n-r-1} \prod_{t \neq \varepsilon}(t - q\varepsilon)$. The head product and the tail product therefore **recombine into a product over all of R**:

$$\varepsilon^r \cdot \varepsilon^{n-r-1} \prod_{v \in R}(v - q\varepsilon) = \varepsilon^{s-1} \prod_{v \in R}(v - q\varepsilon),$$

with no trace of the head selection. This is the exact analogue of step 2 of Rick's Lemma QC, and it is the only step that could have failed.

Step 3: the ratio. The prefactor $\prod_k u_k^{-(N+n)}$ contributes $\varepsilon^{-(N+s)} \prod_{v \in R} v^{-1}$. Collecting, and using $v^{-1}(v - q\varepsilon) = 1 - q\varepsilon/v$ and $\varepsilon^{-N} \prod_k (\varepsilon - q w_k) = \prod_k (1 - q\varepsilon w_k)$,

$$\left.\frac{F_s}{F_{s-1}}\right|_{w_m = \varepsilon} = \prod_{k=1}^{N}(1 - q\varepsilon w_k) \prod_{v \in R}\Big(1 - \frac{q\varepsilon}{v}\Big). \tag{$*$}$$

Step 4: the normaliser does the same. Adjoining $u = \varepsilon$ to $\mathcal{E}^C$ adds, to the numerator, $\prod_{v \in R}(1 - qv/\varepsilon)(1 - q\varepsilon/v) \cdot (1-q)$ from $\prod_{j,k}(1 - q u_k/u_j)$ and $\prod_k (1 - q\varepsilon w_k)$ from $\prod_{j,k}(1 - q u_j w_k)$; and, to the denominator, $\prod_{v \in R}(1 - q\varepsilon v)$ from $\prod_{j<k}(1 - q u_j u_k)$ together with one factor $(1-q)$ from $(1-q)^{n-r}$. The two $(1-q)$'s cancel, and $(1 - qv/\varepsilon) = (1 - q\varepsilon v)$ cancels against the $j < k$ product, leaving

$$\left.\frac{\mathcal{E}_s^C}{\mathcal{E}_{s-1}^C}\right|_{w_m = \varepsilon} = \prod_{k=1}^{N}(1 - q\varepsilon w_k) \prod_{v \in R}\Big(1 - \frac{q\varepsilon}{v}\Big),$$

which is $(*)$. Hence $\widetilde{A}_s / \widetilde{A}_{s-1} = 1$. □

Remark 11 (verification, with a plant). The two ratios of Steps 3 and 4 were checked *separately* against their closed forms — not merely their quotient, which could be 1 with both wrong — over all $(N, \mathbf{M}, m, \mathbf{I}, \varepsilon)$ with $N \leq 3$ and $s - 1 \geq r$: 112/112 for F, 112/112 for $\mathcal{E}^C$, 112/112 for $\widetilde{A}$ itself (`src_proof_check.py`). The proof uses $\varepsilon^2 = 1$ in three places, so the plant is $\varepsilon = 2$: the $\widetilde{A}$ equality must *fail* there, and it fails 24/24. A green at $\varepsilon = 2$ would have meant the instrument was blind.

With multiplier 1 the weight filtration collapses ($W_d = W_0$ for all d), so $\phi_\lambda \phi_\mu$ and the span sit at the same weight and the baseline constants have no boundary-divisor poles — which is what the baseline census shows. A renormalisation has boundary weight $\chi(\varepsilon)$:

$$\Psi_\chi(\mathbf{M})\big|_{w_m = \varepsilon} = \chi(\varepsilon)\, \Psi_\chi(\mathbf{M}^-)\big|_{w_m = \varepsilon},$$

so the condition for $\Psi_\chi\phi_\lambda\phi_\mu$ to stay in the boundary lattice is

$$\textbf{(SR)} \qquad \chi(\varepsilon) = 1 \quad \text{for every } \varepsilon \text{ with } \varepsilon^h = 1.$$

Proposition 12 (boundary lattice lemma)**.** $\mathrm{span}_{\mathcal{O}_{m,\varepsilon}}\{\phi_{\mathbf{I}}\} = \{f : (f(\mathbf{M}) - f(\mathbf{M}^-))|_{S^\varepsilon_m} = 0 \ \forall \mathbf{M} \ni m\}$*, assuming* (SRC) *= Theorem 10. Consequently if* χ *satisfies* (SR) *then every* $\widehat{c}^{\,\nu}_{\lambda\mu}$ *is regular at* S^ε_m*.*

Proof. Verbatim the proof of Proposition 6 with g replaced by 1 and $\mathbf{M}''$ by $\mathbf{M}^-$. The two counts: the defect map onto $k^{\{\mathbf{M}\ni m\}}$ is surjective by the same indicator argument, giving colength 2^{N-1}; and $v_{S^\varepsilon_m}(\widetilde{A}_{\mathbf{I},\mathbf{I}}) = 1$ iff $m \in \mathbf{I}$ — the factor $(1 - w_m^{-2})$, which is the $j = k$ term of (M4)'s $\prod_{j\le k}$ — and no other positive-root factor vanishes on $\{w_m = \varepsilon\}$ at generic q, since $(1 - qw_j^2)$ needs $w_j^2 = q^{-1}$ and the A-type and q-pair factors cut out different divisors. So $v(\det) = \#\{\mathbf{I} \ni m\} = 2^{N-1}$. For the consequence: $f = \Psi_\chi\phi_\lambda\phi_\mu$ has $f(\mathbf{M})|_S = \chi(\varepsilon)\cdot 1 \cdot f(\mathbf{M}^-)|_S = f(\mathbf{M}^-)|_S$. □

Remark 13 (inevitability)*.* The two conditions are met by ψ *for the same reason.* From `eq:dict`, $\rho^X - 1 = q(t-1)^2/((1-qt)(1-q/t))$ with $t = w_a^h$, so $\{\rho^X = 1\} = \{q = 0\} \cup \{w_a^h = 1\}$: the obstruction vanishes exactly on the h-th roots of unity. And $\psi(v) = (1 - qv^h)/(1-q)$ satisfies $\psi(\varepsilon) = 1$ exactly when $\varepsilon^h = 1$. **The locus where the multiplier is already trivial is precisely the locus where the renormaliser is required to be trivial.** The boundary divisors of $\det \widetilde{A}$ are $\{w_m^h = 1\}$ — the same set. I did not arrange this; I ran an arm that broke, and the diagnosis predicted that it must break at $\varepsilon = -1$ and *not* at $\varepsilon = +1$, which is what the data show.

8 Theorem C: the exact family

Theorem 14 (exact characterisation)**.** *Let* $\chi \in \mathbb{C}(q)(v)^\times$ *be regular and non-vanishing at the generic point of each divisor below,* $\Psi = \Psi_\chi$*, and put* $\kappa := \chi(x)\chi(1/x) \in k_{ab}^\times$*. Fix* λ, μ*.*

1. (pair divisors) *For* $a < b$*, the following are equivalent:*
   - *(a) every* $\widehat{c}^{\,\nu}_{\lambda\mu}$ *is regular at* H_{ab}*;*
   - *(b)* $(\kappa g - 1)\cdot(\phi_\lambda\phi_\mu)(\mathbf{M}'')|_H = 0$ *for every* $\mathbf{M}'' \subseteq [N]\setminus\{a,b\}$*;*
   - *(c)* $\kappa = g^{-1}$*, or no* $\mathbf{M}'' \subseteq [N]\setminus\{a,b\}$ *satisfies both* $\mathbf{M}'' \ge \lambda$ *and* $\mathbf{M}'' \ge \mu$*.*
2. (boundary divisors) *For* $\varepsilon^h = 1$*, every* $\widehat{c}^{\,\nu}_{\lambda\mu}$ *is regular at* S^ε_m *if* $\chi(\varepsilon) = 1$*; and if* $\chi(\varepsilon) \ne 1$ *it is regular there only when* $(\phi_\lambda\phi_\mu)(\mathbf{M}^-)|_S = 0$ *for all* $\mathbf{M} \ni m$*.*

Hence the renormalisations that are pole-free on all *these divisors for* all (λ, μ) *are exactly*

$$\boxed{\chi = \psi\cdot\eta, \qquad \eta(x)\eta(1/x) = 1, \qquad \eta(\varepsilon) = 1 \ \ \forall\, \varepsilon^h = 1.}$$

Proof. (1) Put $f := \Psi\phi_\lambda\phi_\mu$. By (QC) and the computation in Theorem 8, $f(\mathbf{M})|_H = \kappa g^2 f(\mathbf{M}'')|_H$, so the defect is $D_{\mathbf{M}}(f) = (\kappa g^2 - g)f(\mathbf{M}'')|_H = g(\kappa g - 1)f(\mathbf{M}'')|_H$. By Proposition 6, (a) $\iff$ $D_{\mathbf{M}}(f) = 0$ for all $\mathbf{M}$, and $g \ne 0$ gives (b). For (b) $\iff$ (c): if $\kappa = g^{-1}$ then $\kappa g - 1 = 0$. Otherwise $\kappa g - 1 \ne 0$ in k_{ab}, and since $\Psi(\mathbf{M}'')|_H \ne 0$ by hypothesis we need $(\phi_\lambda\phi_\mu)(\mathbf{M}'')|_H = 0$ for all $\mathbf{M}''$. That holds iff no $\mathbf{M}'' \ge \lambda, \mu$: "$\Leftarrow$" is (M3); "$\Rightarrow$" is Rick's Lemma L (his §2bis: a degeneration $w_c = t^{-e_c}$, $t \to 0^+$, in which $\phi^0_\kappa(\mathbf{M}'') \to 1$ when $\mathbf{M}'' \ge \kappa$), which shows $\phi^0_\lambda\phi^0_\mu(\mathbf{M}'')|_H \ne 0$ at $q = 0$ and hence $\ne 0$ at generic q.

(2) is Theorem 10 and Proposition 12, with the same defect computation at $g = 1$.

For the final display: (COB) is $\kappa = g^{-1}$, and ψ is one solution by Theorem 3, so the solution set of (COB) is the coset $\psi\cdot\{\eta : \eta(x)\eta(1/x) = 1\}$; (SR) is $\chi(\varepsilon) = 1$, and $\psi(\varepsilon) = 1$ for $\varepsilon^h = 1$ (Remark 13), so it transfers to $\eta(\varepsilon) = 1$. □

Corollary 15 (the brief's Theorem C, answered negatively)**.** *The pair-divisor pole order is* ***not*** *invariant across the family of diagonal renormalisations. By Proposition 6 it is at most* 1*, and by Theorem 14(1) it is* 0 *on the coset* $\psi \cdot \{\eta\}$ *and* 1 *off it (on every non-degenerate* (λ, μ)*). The brief asked for an invariance theorem that would bind Rick's unwritten geometric normalisation; instead the pole-removing normalisations are pinned down exactly, and any geometric normalisation that is diagonal in this sense* must *lie in the boxed coset.*

Remark 16 (monomial twists). $\eta(v) = v^k$ satisfies $\eta(x)\eta(1/x) = 1$ for every k, and $\eta(\varepsilon) = \varepsilon^k$, so in type C ($h = 2$, $\varepsilon = \pm 1$) the admissible monomial twists are exactly the *even* powers. This was used as a falsifiable prediction: $\chi = v\psi(v)$ must fail at $w = -1$ only, and $\chi = v^2\psi(v)$ must be clean. See §9.

9 Verification

All computations: pure SymPy, engine `proofs/code-1017c2-coboundary/q484_engine.py` (the 2026-10-17 copy, the only one carrying $\mathcal{E}^B$; for type C it is byte-identical in the relevant code path to the copies used on 10-15 and 10-16, checked by `diff`). Census: every (λ, μ, ν) with $\mu \geq \lambda$ in the linear extension and $c^{\nu}_{\lambda\mu} \neq 0$; $N = 3$, type C; 115 non-zero constants. A constant is counted *non-Laurent* iff, after `cancel`, its denominator has a non-monomial irreducible factor.

Instrument controls.

- **Baseline reproduces.** $\chi = 1$ at symbolic q returns 13/115 non-Laurent, which is the 2026-10-16 census number, and the poles are on pair divisors only.
- **The q-specialisation is faithful.** $\chi = 1$ at $q = 3/7$ returns 13/115 *and the symmetric difference against the symbolic run is empty* — the same 13 triples with the same denominators, not merely the same count. This licenses reading the other arms at $q = 3/7$.
- **The instrument can see movement in both directions**: one arm removes poles (ψ), one arm adds them ($v\psi$). A green that could only ever be green is not evidence.

Arms. Each χ gives $\Psi_\chi(\mathbf{M}) = \prod_{m \in \mathbf{M}} \chi(w_m)$ and a full recomputation of all 115 constants from $\widehat{A}$ by the triangular solve — an independent route to $\widehat{c}$, not the $\ell_\nu(\Psi\phi_\lambda\phi_\mu)$ rewriting used in the proof. Predictions were written to `PREDICTIONS.md` before the arms finished.

arm	$\chi(v)$	(COB)	(SR)	result
`id`	1	fail	holds	13/115; pair divisors only
`psi`	$(1 - qv^2)/(1 - q)$	holds	holds	**0/115**
`x2psi`	$v^2(1 - qv^2)/(1 - q)$	holds	holds	**0/115**
`xpsi`	$v(1 - qv^2)/(1 - q)$	holds	fails at -1	57/115; **zero** pair-divisor, all at $w_i + 1$
`x`	v	fail	fails at -1	57/115; pair-divisor *and* $w_i + 1$
`sq`	$((1 - qv^2)/(1 - q))^2$	fail	holds	13/115; **the same 13, same denominators**
`h1`	$(1 - qv)/(1 - q)$	fail	fails at -1	57/115; pair-divisor *and* $w_i + 1$
`unrm`	$1 - qv^2$	fail	fails at ± 1	57/115; pair-divisor, $w_i + 1$ *and* $w_i - 1$

Every pre-registered prediction holds, at divisor level. The table's "result" column is not a count: each entry records *which divisor classes* carry poles. Reading the two condition columns against the three divisor classes:

- **(COB) governs the pair divisors and nothing else.** The four arms with (COB) failing (`id`, `x`, `sq`, `h1`, `unrm`) all have pair-divisor poles; the three with (COB) holding (`psi`, `x2psi`, `xpsi`) have *none*, including `xpsi`, which is riddled with poles of the other kind.
- **(SR) governs the boundary divisors, and separately at each ε.** The arms with $\chi(-1) \neq 1$ (`xpsi`, `x`, `h1`, `unrm`) have poles at $w_i + 1$ and no others have any; `unrm` is the *only* arm with $\chi(+1) \neq 1$ and the *only* arm, of eight, that produces a $w_i - 1$ pole.
- **The positive prediction was made before it was measured.** $\chi = v^2\psi$ satisfies both conditions and was predicted clean at a value of χ not previously run; it returns 0/115.
- **The sharpest set prediction holds.** Theorem 14(1) makes the pair-divisor pole locus depend on χ only through the predicate $\kappa g = 1$, which fails for both `id` and `sq`; so `sq` must return not merely 13 but *the same* 13 triples. It does, with identical denominators on every one.

A grading lesson, paid for in this session. `x` and `xpsi` both return 57, and their sets of 57 triples are **identical** — symmetric difference empty. Grading by count would have called them the same; grading by set, which is my standing rule (`two-criteria-can-fit-the-same-count-and-differ-as-sets`), would *also* have called them the same. They are not: on exactly the 13 baseline triples their denominators differ, `x` retaining the pair-divisor factor that `xpsi` has shed, e.g. at $(\emptyset, \emptyset, \{1, 2\})$

$$\texttt{x}: \ (w_2+1)(w_1+1)(w_1w_2-1) \qquad \text{versus} \qquad \texttt{xpsi}: \ (w_2+1)(w_1+1).$$

The whole of Theorem 8 lives in that difference. So: *a set is itself a projection. When two arms agree as sets, descend to the label each element carries* — here, the pole divisor, not the triple. One level below the count-to-set rule, and the first time it has mattered.

Generic q. The main positive arm was also run at symbolic q rather than $q = 3/7$: $\chi = \psi$ over $\mathbb{C}(q, w_1, w_2, w_3)$ returns **0/115** as well, so Theorem 8 is confirmed at generic q and not only at a specialisation. The baseline was likewise run at symbolic q (13/115). The other six arms were read at $q = 3/7$, licensed by the empty symmetric difference of the two baseline runs.

10 Where the pre-registered mechanism broke

The brief's Theorem-B argument reads $\prod_{m \in \mathbf{M}} \psi(w_m)$ as a *diagonal rescaling of the basis*, and concludes that a structure constant is multiplied by a ratio of ψ's — a unit at H_{ab}, hence unable to cancel a pole. That conclusion is correct for the operation it names, and the operation it names is not the one in the formula.

- **Label-indexed** rescaling: $\phi_{\mathbf{I}} \mapsto \chi_{\mathbf{I}}\phi_{\mathbf{I}}$ with $\chi_{\mathbf{I}} \in F$. Then $\widehat{c}^{\nu}_{\lambda\mu} = (\chi_\lambda\chi_\mu/\chi_\nu)\, c^{\nu}_{\lambda\mu}$, a unit multiple; poles cannot move. *True.*
- **Point-indexed** rescaling: $\phi_{\mathbf{I}} \mapsto \Psi\phi_{\mathbf{I}}$ with $\Psi \in \mathcal{A}$ a single function of the *point*. Then $\widehat{c}^{\nu}_{\lambda\mu} = \ell_\nu(\Psi\, \phi_\lambda\phi_\mu)$, which is not a unit multiple of $c^{\nu}_{\lambda\mu}$ at all: multiplication by a fixed element of the algebra reshuffles coordinates.

The product $\prod_{m\in\mathbf{M}}$ is indexed by $\mathbf{M}$, the point. And the decisive point is that **Theorem A requires the point-indexed reading**: the collapse $\mathbf{M} \mapsto \mathbf{M}''$ acts on the point, so a label-indexed $\chi_{\mathbf{I}}$ cancels out of the collapse ratio entirely and changes no multiplier. So the brief proved Theorem A under reading 2 and refuted Theorem B under reading 1, in adjacent paragraphs, with the same symbol.

> **Rule.** The index set of a rescaling decides whether it is a change of basis or multiplication by an algebra element. Before reasoning about what a rescaling does to structure constants, say out loud which set the product runs over — and check that the *other* theorems in the same document are using the same reading.

This is a relative of the one-coordinate-match family (`the-missing-coordinate-is-one-rule-in-nine-drawe`) but it is a different animal and I want the distinction recorded. There, two *objects* agreed on one coordinate and I inferred identity. Here there is one object and two *readings of my own notation*, and the brief's two theorems required *incompatible* readings: the document was internally inconsistent while each paragraph was locally correct. A consistency check across paragraphs of one's own brief would have caught it; a check of any single paragraph would not.

11 Consequences

Retraction 1. *Q491's criterion — $c^{\nu}_{\lambda\mu}$ Laurent $\iff$ $|\nu| - \max(|\lambda|, |\mu|) \le 1$, measured* 225/225 *with zero false positives and zero false negatives — is a property of the $\mathcal{E}^C$-normalised basis $\widetilde{A}$, not of the algebra $\mathcal{A}$. At $N = 3$ every non-Laurent constant has its poles on pair divisors only, so under the ψ renormalisation all* 115 *become Laurent and the criterion becomes vacuous. The criterion remains true and useful* of *$\widetilde{A}$; what is withdrawn is any reading of it as intrinsic.*

Retraction 2. *The negative half of Q500 — "pole $\iff$ Pfaffian category", with the pole as an obstruction — is withdrawn in the form I stated it. The pole is not an invariant of the algebra; it is an invariant of a choice of normaliser, and the choice that removes it is explicitly available. The identity $\rho^X - 1 = q\, w_a^h([K^X(w_a)]_0^1)^2$ is unaffected: it is a true statement about the* cochain *ρ.*

What replaces them is better, and is the actual content of today:

> The obstruction is a **cohomology class**, and it is **zero**. The poles are removable *because* ρ splits as $\varphi(w_a)\varphi(w_b)$. Had it not split — had it depended on a third w_c, or failed to factor by variable — Theorem 14 says no point-diagonal renormalisation could have removed the poles, and the obstruction would have been real. The non-vanishing of $\rho - 1$ and the existence of the poles are both statements about a representing cochain. What I had been calling the obstruction was the cochain.

This also sharpens Rick's sentence. He wrote that my poles *measure the normalisation defect*. Strictly that identifies two objects — the defect of the *basis* ($\rho \neq 1$) and the poles of the *structure constants* — and my pre-registered objection to the identification was the right instinct applied to the wrong case: they are different objects, and here they happen to be controlled by the same cochain, so trivialising one trivialises the other. The correct statement is the implication, not the identity: *ρ a coboundary $\Rightarrow$ the structure-constant poles are removable*, with Theorem 14 supplying the converse.

12 Gaps and questions

Gap 1 (type B, and $N \geq 4$)**.** *Theorems 3–14 are stated for general h and use only (QC), (M3), (M4) and (SRC); the type B input (QC) is proved (`thm:rhoB`). But the* census *is type C, $N = 3$ only. Q502 (re-measure every type B number against $\mathcal{E}^B$) is untouched. At $N = 4$ Rick reports that the first-order route behaves differently (his (b) and (c) both fail first at $N = 4$); nothing in §6 is first-order, so I expect Theorem 8 to survive, but it is unmeasured.*

Gap 2 (other divisors)**.** $\det \widetilde{A}$ *also vanishes on the A-type divisors $\{w_i = w_j\}$ and on q-divisors $\{q \cdot \text{monomial} = 1\}$. No baseline constant has a pole there, and no arm produced one, but I have no theorem.*

New questions. **Q510** — *is $\widehat{A}$ the weight function of a model?* $\widehat{A}$ is a basis with honest GKM collapse at both the pair and the boundary divisors and a genuine multiplication rule with no poles on them. Rick proposes it should be Koncki's twisted motivic Chern class for $LG(n, 2n)$. The test is cheap and sharp: $\psi|_{q=0} = 1$, so $\widehat{A}|_{q=0} = \widetilde{A}|_{q=0}$, and the two can only differ in their q-deformation; compare $\widehat{A}$ against Koncki's table for $LG(2, 4)$ on the entry Rick predicts is the odd one out, $(\{1, 2\}, \emptyset)$. He states he has not checked Koncki's table; neither have I. Held at `agent-summary`; a first-hand read is the prerequisite.

Gate Q504 observed. No novelty sentence is made here about Q484, Q491 or Q492. Digalaki–Wheeler, *Structure Constants for Free-Fermion Six Vertex Model Symmetric Functions* (Mittag-Leffler, July 2026, in preparation, no arXiv ID) is held only at `agent-summary`, and their hypothesis is the free-fermion locus, which I conjecture is my $\rho = 1$. Theorems 3, 8 and 14 concern my own normaliser and the removability of my own poles; Retractions 1 and 2 are retractions of my own claims. Neither is a priority claim.

Acknowledgement

Lemma QC, Proposition 5 and Theorem 6 of `2026-10-17-q488-converse-order-q1-proofs.pdf` are Rick's. The lattice lemma above is his Proposition 5, re-derived here with its two numerical inputs checked independently against my engine; Theorem 14(1)(c) uses his Lemma L. The hunch that the poles are a normalisation artefact is his, it was flagged by him as a hunch, I pre-registered against it, and he was right.